in the magnitude of ($N - Z$) means an increasing N and a diminishing Z. Those nuclei which lie to the right of the lowest point, have a greater number of neutrons and a smaller number of protons than the stable nucleus, and will gradually change into the stable nucleus by the emission of one or more electrons. On the other hand, the nuclei situated to the left, which contain more protons and fewer neutrons than the stable nucleus, will effect their change by the emission of positrons or by the capture of an electron from the extranuclear atomic structure (K-capture), as shown by the arrows in Figure 18. All this is in full agreement with experimental evidence. Figures 18 and 19 show the conditions for the specific mass numbers 91 and 92.

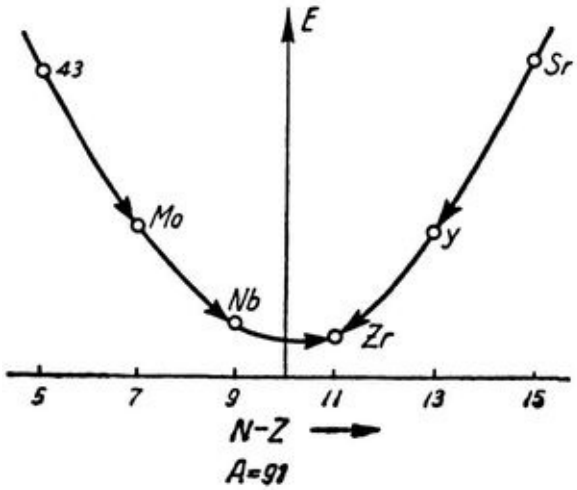


Figure 18.—Binding energy of uneven nuclei.

The situation is totally different in the case of ‘even’ atoms, i.e. where $N + Z$ is an even number. Here one of the finer details of the energy surface (not reflected in the equation formulated in the fourth chapter of this book) shows up, namely: the difference already discussed in the stability, and consequently, in the binding energies, of the ‘doubly even’ and ‘doubly odd’ nuclei. As already pointed out, the former possess a considerably greater degree of stability than the latter. Therefore, in order to represent both nuclear types, we must draw two different curves; the first one, representing the ‘doubly even’—and therefore the more stable—nuclei lies below the other one which represents the ‘doubly odd’ nuclei. The change of one nucleus into another can take place by discrete steps only, by the emission of one electron or positron (plus the necessary neutrino) at one time and never by a simultaneous emission of two electrons or two positrons. While it is still possible, for instance, for the nucleus in which $N - Z = 6$ (e.g. $N = 49$ and $Z = 43$—a ‘doubly odd’ nucleus) to change into the ‘doubly even’ Mo nucleus in which $N - Z = 8$ ($N = 50$ and $Z = 42$), by the emission of a positron, as shown by the arrow in Figure 19, for energy is liberated in this process. But in order